

Kubernetes Pods, Deployments and Scaling — Notes

Overview

This note explains core Kubernetes concepts using a simple analogy and practical commands:

- Pods
- ReplicaSets
- Deployments
- Scaling
- Labels and Selectors

Core Concepts

Pod

- Smallest unit in Kubernetes
- Runs one or more containers
- Represents a single instance of an application

Analogy:

A Pod is like a single chef in a restaurant kitchen preparing food.

ReplicaSet

- Ensures a specified number of Pods are always running
- Automatically replaces failed Pods

Analogy:

If one chef leaves, the manager hires another to maintain the required number of chefs.

Deployment

- Manages ReplicaSets and Pods
- Provides updates, rollbacks, and scaling

Analogy:

The restaurant manager who decides:

- How many chefs are needed
- When to hire more
- When to replace them

YAML Example (Pod Creation)

```
apiVersion: v1
kind: Pod
metadata:
  name: mypod
  labels:
    app: myapp
spec:
  containers:
    - name: nginx-container
      image: nginx
```

Apply the Pod:

```
kubectl apply -f pod.yaml
```

Deployment Example

```
kubectl create deployment myapp --image=nginx
```

Check deployment:

```
kubectl get deployments
```

Scaling Applications

Scale up:

```
kubectl scale deployment myapp --replicas=3
```

Scale down:

```
kubectl scale deployment myapp --replicas=1
```

Labels and Selectors

Labels

- Key-value pairs attached to resources

Example:

```
labels:  
  app: myapp  
  env: production
```

Selectors

- Used to find matching resources based on labels

Example:

```
kubectl get pods -l app=myapp
```

Useful kubectl Commands

Check Pods:

```
kubectl get pods
```

Detailed info:

```
kubectl describe pod <pod-name>
```

Logs:

```
kubectl logs <pod-name>
```

Delete resource:

```
kubectl delete pod <pod-name>
```

Delete deployment:

```
kubectl delete deployment myapp
```

Key Understanding

- Pod = single running instance
- ReplicaSet = maintains number of Pods
- Deployment = manages everything above
- Scaling = increasing or decreasing Pods
- Labels = identification tags
- Selectors = filtering mechanism

Summary

- Kubernetes manages applications through Pods
- Deployments automate scaling and updates
- ReplicaSets ensure availability
- Scaling allows handling traffic changes
- Labels and selectors help organize and manage resources

Goal

Be able to:

- Create Pods using YAML
- Deploy applications
- Scale applications up and down
- Use kubectl for managing resources
- Understand how Kubernetes maintains application availability